Date

PRIVACY-ENHANCED SYNTHETIC DATA GENERATION FOR AUTONOMOUS VEHICLE TRAINING

14th - 28th Feb

Project Id :- R26-CS-010

Student Id :- IT22541284

Name :- Yubitha S

Individual Component : Generate realistic, privacy aware synthetic autonomous vehicle datasets.

Research Topic ⇒ Privacy-Enhanced Synthetic Data Generation for Autonomous Vehicle Training.

Summary of RP work carried out during the reporting period

Date	RP work
14/02/2026	Read research papers on autonomous vehicle simulation (AirSim Simulator) and why simulation is used for sensor data generation and testing.
19/02/2026	Reviewed papers/articles on sensor modelling in simulators (camera and LiDR data characteristics, noise/limitations).
20/02/2026	Looked into research on radar/distance sensing simulation and how it is approximated in platforms like AirSim.
23/02/2026	Installed basic environment requirements for simulator work. Installed AirSim.
25/02/2026	Consolidated literature findings into a short summary for methodology.

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Date

14th Feb - 28th Feb

Supervisor's comments.

Student progressing well in research project : Yes / NO

If NO : Still in a initial stage.

Project work to be carried out before next evaluation period :

Signature :

Date : 25/02/2026

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Date

Feb 28th - 14th March.

PRIVACY - ENHANCED SYNTHETIC DATA GENERATION FOR AUTONOMOUS VEHICLE TRAINING

Project ID - R26-CS-010

Student ID - IT22541284

Name - Yubitha S

Individual Component - Generate realistic, privacy aware systhetic autonomous vehicle datasets, with security features.

Summary of RP work carried out during the reporting period.

Date	RP Work.
28/02/2026	Read research papers on autonomous vehicle Simulator (CARLA) and why simulation is used for sensor data generation and testing.
01/02/2026	Reviewed papers and articles on sensor modelling in simulators camera and LIDR data characteristics, noise)
03/03/2026	Looked into research on radar/distance sensing simulation and how it is approximated in platform CARLA.
05/03/2026	Installed basic environment requirements for simulator work. Installed CARLA simulator.
11/03/2026	Consolidated literature findings into a short summary for methodology.

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Date

28th Feb - 14th March

Supervisor's comments :-

Student progressing well in research project :- Yes / No

IF No :-

P.

Project work to be carried out before next ev evaluation period :-

Code base checked :- Yes / No

Signature :-

Date :- 17/05/2026

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20th March – 27th April

Date

PRIVACY-ENHANCED SYNTHETIC DATA GENERATION FOR AUTONOMOUS VEHICLE TRINING

Project ID ⇒ R26-CS-010 Student ID ⇒ IT22541284 Name ⇒ Yubitha S.

Individual component ⇒ Generate realistic, privacy aware systhetic autonomous vehicle datasets, (with Encryption) Security Features.

Summary of RP work carried out during the reposting period.

Date	RP work
20/03/2026	Finalised CARLA Simulator environment setup. Verified python API connection and confirmed sensor modules (RGB Camera, depth, segmentation) are accessible and returning valid outputs.
25/03/2026	Designed the RBAC policy structure for simulation outputs. Defined three roles: Simulation engineer, Privacy engineer, ML Researcher. Drafted role permission matrix and began implementation of Credential check wrapper.
01/04/2026	Implemented SHA-256 cryptographic hashing module. Integrated hash generation immediately upon dataset batch capture. Created hash manifest file template for integrity logging alongside each batch
07/04/2026	Implemented AES-256 encryption in CBC mode for generated datasets batches. Configured key management to be stored seperately from dataset storage. Tested encryption and decryption on sample RGB images batches.
14/04/2026	Ran initial CARLA simulation using configured scenes. Captured RGB images, depth maps and segmentation data across 3 driving scenarios. Applied SHA-256 hashing and AES-256 encryption to all captured batches successfully.
21/04/2026	Audited RBAC access control from simulation run. Verified that role credential checks are enforced correctly at each pipline stage. Identified and resolved one access control bypass edge case in the API wrapper.
27/04/2026	Began Validating dataset format and sensor coverage across all generated batches. Cross-checked hash manifests to confirm data integrity. Prepared summary notes for cybersecurity layer design documentation

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Date

20th March – 27th April.

Supervisor's comment :-

Student progressing well in research project :- (Yes) / No

If No :-

Project work to be carried out before next evaluation period :-

Signature :-

Date :- 29/04/2026

innovate

Date

PRIVACY-ENHANCED SYNTHETIC DATA GENERATION FOR AUTONOMOUS VEHICLE TRAINING

1st May – 15th May

Project ID:- R26-CS-010 Name:- Yubitha S

Student ID:- IT 22541284

Individual component :- Generate realistic, privacy away synthetic autonomous vehicle datasets with security feature.

Summary of RP work carried out during the reporting period.

Date	RP work.
01/05/2026	Completed full-scale CARLA simulation run across 5 diverse driving scenarios. Captured RGB images, deapth maps for each scenario batch successfully.
03/05/2026	Applies SHA-256 cryptographic hashing to all 5 dataset batches from the simulation run. Verified hash manifest files were correctly generated alongside each batch.
07/05/2026	Applied AES-256 encryption (CBC mode) to all validated dataset batches. Confirmed encrypted files stored seperately from key management files to prevent unauthorized access.
10/05/2026	Enforced RBAC credential checks across all pipeline stages during batch processing. Logged access events for each role (simulation engineer, Privacy engineer, ML Researcher)
13/05/2026	Reviewed RBAC access logs for anomalies. All role-based access restrictions were correctly enforced. No unauthorized access attemps detected during May 1 simulation run.
15/05/2026	Completed first fortnight review. Summarized simulation run results, confirmed all hash manifests intact. No integrity mismatches found across any generated batch

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Date _____

1st May - 15th May.

Supervisor's comment :-

Student Progressing well in research project :- YES / NO

If NO :-

Project work to be carried out before next evaluation period :-

Signature :-

Date :-

innovate

Date

1st June – 15th June 2026

PRIVACY – ENHANCED SYNTHETIC DATA GENERATION FOR AUTONOMOUS VEHICLE TRAINING

Project ID – R26-CS-010

Student ID – IT22541284

Name – Yubitha S

Individual Component – Generate realistic, privacy aware Synthetic autonomous vehicle datasets, with security features.

Summary of RP work carried out during the reporting period.

Date	RP Work
01/06/2026	Began comprehensive validation of sensor coverage across all generated dataset batches. Checked that RGB, depth and segmentation outputs are consistent and complete.
04/06/2026	Identified two batches with incomplete segmentation data due to CARLA sensor configuration mismatch. Documented issue and planned corrective simulation re-run.
07/06/2026	Reconfigured CARLA sensor parameters to resolve segmentation data gaps. Re-ran affected simulation scenarios and captured corrected dataset batches.
10/06/2026	Re-validated corrected batches. Applied SHA-256 hashing and AES-256 encryption to all regenerated data. Updated hash manifests and confirmed RBAC logs are complete.
13/06/2026	Cross-checked all batch sensor outputs against expected coverage specifications. Confirmed all driving scenarios now have full RGB, depth and segmentation data.
15/06/2026	Completed first fortnight sensor validation summary. All batches confirmed complete and integrity-verified. Ready to proceed with dataset handover preparation.

Date

Supervisor's Comments :-

Student progressing well in research project :- Yes / No

If No :-

Project work to be carried out before next evaluation period :-

Code base checked :- Yes / No



Signature :-

Date :- 17/6/26

Date

16th June – 30th June 2026

PRIVACY – ENHANCED SYNTHETIC DATA GENERATION FOR AUTONOMOUS VEHICLE TRAINING

Project ID – R26-CS-010

Student ID – IT22541284

Name – Yubitha S

Individual Component – Generate realistic, privacy aware Synthetic autonomous vehicle datasets, with security features.

Summary of RP work carried out during the reporting period.

Date	RP Work
16/06/2026	Performed full RBAC audit log review covering all simulation runs from May and June. Verified role-based access restrictions enforced correctly across all pipeline stages.
19/06/2026	Organized final dataset directory structure for handover. Grouped batches by scenario type and documented metadata including scenario parameters and simulation configurations.
22/06/2026	Prepared final encrypted and integrity-verified synthetic dataset package for handover to IT22110220 (Utility Evaluation member). Included hash manifests and RBAC audit logs.
25/06/2026	Delivered final synthetic dataset package to IT22110220. Confirmed receipt and dataset usability with team member. Resolved minor file format query raised by IT22110220.
27/06/2026	Drafted initial cybersecurity layer design documentation covering SHA-256 integrity verification, AES-256 encryption and RBAC implementation for the group thesis.
30/06/2026	Completed June documentation draft. Submitted draft section to supervisor for early review. Prepared structure outline for July thesis writing phase.

Date

16th June – 30th June 2026

Supervisor's Comments :-

Student progressing well in research project :- Yes / No

If No :-

Project work to be carried out before next evaluation period :-

Code base checked :- Yes / No

Signature :-

Date :- 1/7/26

Date

1st July – 15th July 2026

PRIVACY – ENHANCED SYNTHETIC DATA GENERATION FOR AUTONOMOUS VEHICLE TRAINING

Project ID – R26-CS-010

Student ID – IT22541284

Name – Yubitha S

Individual Component – Generate realistic, privacy aware Synthetic autonomous vehicle datasets, with security features.

Summary of RP work carried out during the reporting period.

Date	RP Work
01/07/2026	Began detailed thesis writing for the cybersecurity layer section. Structured content around SHA-256, AES-256 and RBAC subsections with supporting diagrams and pseudocode.
04/07/2026	Documented the SHA-256 hash verification workflow in full. Included flow diagrams showing pre-capture and post-capture integrity checks and hash manifest generation steps.
07/07/2026	Documented AES-256 encryption implementation including key management approach, CBC mode configuration and integration points within the CARLA API pipeline wrapper.
10/07/2026	Documented RBAC policy design: role definitions, permission matrices and enforcement mechanism within the pipeline. Included sample RBAC audit log format and entries.
13/07/2026	Reviewed membership inference attack (MIA) test results from IT22066916. Analysed findings to assess re-identification risk in the generated synthetic dataset.
15/07/2026	Documented MIA findings in thesis security section. Discussed implications for pipeline security and noted proposed mitigation strategies based on attack results.

Date

1st July – 15th July 2026

Supervisor's Comments :-

Student progressing well in research project :- Yes / No

If No :-

Project work to be carried out before next evaluation period :-

Code base checked :- Yes / No



Signature :-

Date :- 18/7/26

Date

16th July – 31st July 2026

PRIVACY – ENHANCED SYNTHETIC DATA GENERATION FOR AUTONOMOUS VEHICLE TRAINING

Project ID – R26-CS-010

Student ID – IT22541284

Name – Yubitha S

Individual Component – Generate realistic, privacy aware Synthetic autonomous vehicle datasets, with security features.

Summary of RP work carried out during the reporting period.

Date	RP Work
16/07/2026	Reviewed utility evaluation results from IT22110220 including mAP and FID scores for models trained on the synthetic dataset. Analysed trade-off between security and utility.
19/07/2026	Incorporated utility trade-off analysis into the thesis performance section. Compared results against non-private baseline as referenced in the evaluation framework.
22/07/2026	Collaborated with IT22309556 (Privacy Integration) to review epsilon-DP results. Cross-referenced with security layer outcomes for combined privacy-security discussion in thesis.
25/07/2026	Completed first full draft of the cybersecurity layer chapter for the group thesis. Submitted draft chapter to supervisor for review and feedback.
28/07/2026	Received initial supervisor feedback on thesis draft chapter. Noted revisions required for threat model discussion and pipeline security justification sections.
31/07/2026	Began implementing supervisor feedback revisions. Strengthened threat model discussion and expanded pipeline-level security justifications with references to reviewed literature.

Date

16th July – 31st July 2026

Supervisor's Comments :-

Student progressing well in research project :- Yes / No

If No :-

Project work to be carried out before next evaluation period :-

Code base checked :- Yes / No



Signature :-

Date :- 3/8/26

Date

1st August – 15th August 2026

PRIVACY – ENHANCED SYNTHETIC DATA GENERATION FOR AUTONOMOUS VEHICLE TRAINING

Project ID – R26-CS-010

Student ID – IT22541284

Name – Yubitha S

Individual Component – Generate realistic, privacy aware Synthetic autonomous vehicle datasets, with security features.

Summary of RP work carried out during the reporting period.

Date	RP Work
01/08/2026	Completed supervisor feedback revisions to the cybersecurity layer thesis section. Improved threat model narrative and added stronger citations from reviewed AV security literature.
04/08/2026	Conducted final QA review of all synthetic dataset batches. Verified all hash manifests are intact and no integrity mismatches present across the full dataset collection.
07/08/2026	Re-confirmed AES-256 encryption keys are securely stored and separate from dataset storage. Re-ran RBAC audit to ensure no access violations across all stored batches.
10/08/2026	Collaborated with group members on integrating individual thesis chapters. Reviewed privacy integration (IT22309556) and compliance sections (IT22066916) for consistency.
13/08/2026	Prepared demonstration materials for Progress Presentation 2 (17–19 Aug milestone). Focused on showcasing the full secure data generation pipeline and RBAC audit log evidence.
15/08/2026	Finalised Progress Presentation 2 slides. Rehearsed pipeline walkthrough and prepared responses to anticipated examiner questions on security controls and dataset validation.

Date

1st August – 15th August 2026

Supervisor's Comments :-

Student progressing well in research project :- Yes / No

If No :-

Project work to be carried out before next evaluation period :-

Code base checked :- Yes / No



Signature :-

Date :- 15/8/26

Date

16th August – 31st August 2026

PRIVACY – ENHANCED SYNTHETIC DATA GENERATION FOR AUTONOMOUS VEHICLE TRAINING

Project ID – R26-CS-010

Student ID – IT22541284

Name – Yubitha S

Individual Component – Generate realistic, privacy aware Synthetic autonomous vehicle datasets, with security features.

Summary of RP work carried out during the reporting period.

Date	RP Work
17/08/2026	Delivered Progress Presentation 2. Presented secure synthetic data generation component, pipeline architecture, SHA-256 and AES-256 implementation and RBAC audit results.
20/08/2026	Documented examiner feedback from Progress Presentation 2. Key feedback included expanding scalability discussion and clarifying key management approach in the thesis.
23/08/2026	Incorporated Progress Presentation 2 feedback into the thesis draft. Expanded scalability and security considerations section with additional technical detail and justification.
26/08/2026	Reviewed and finalised the complete cybersecurity chapter for submission to the group thesis consolidation. Confirmed all figures, tables and references are correctly formatted.
29/08/2026	Submitted finalised cybersecurity chapter to the group for consolidation into the final report draft. Began proofreading the combined group thesis document.
31/08/2026	Completed group thesis proofreading for cybersecurity and methodology sections. Submitted consolidated feedback to group lead for final report preparation.

Date

16th August – 31st August 2026

Supervisor's Comments :-

Student progressing well in research project :- Yes / No

If No :-

Project work to be carried out before next evaluation period :-

Code base checked :- Yes / No



Signature :-

Date :- 2/9/26

Date

1st September – 15th September 2026

PRIVACY – ENHANCED SYNTHETIC DATA GENERATION FOR AUTONOMOUS VEHICLE TRAINING

Project ID – R26-CS-010

Student ID – IT22541284

Name – Yubitha S

Individual Component – Generate realistic, privacy aware Synthetic autonomous vehicle datasets, with security features.

Summary of RP work carried out during the reporting period.

Date	RP Work
01/09/2026	Began final proofreading and formatting of the group thesis. Reviewed references, figure captions and table formatting across the cybersecurity and methodology sections.
04/09/2026	Finalised work breakdown structure and contribution summary for thesis appendix. Confirmed all individual deliverables are complete: dataset batches, hash logs, RBAC audit logs.
07/09/2026	Began drafting the research paper summarising the privacyenhanced synthetic data generation framework. Structured abstract, introduction and methodology sections.
10/09/2026	Drafted results and discussion sections of the research paper. Included security validation outcomes, hash test results and RBAC access control audit findings.
13/09/2026	Prepared final viva presentation slides (12–14 Oct milestone). Structured demonstration of the complete secure pipeline from CARLA simulation through to encrypted dataset delivery.
15/09/2026	Completed first full draft of the research paper. Submitted to supervisor for feedback ahead of the 15 Oct submission deadline. Noted revision areas flagged by supervisor.

Date

1st September – 15th September 2026

Supervisor's Comments :-

Student progressing well in research project :- Yes / No

If No :-

Project work to be carried out before next evaluation period :-

Code base checked :- Yes / No



Signature :-

Date :- 19/9/26

Date

16th September – 15st October 2026

PRIVACY – ENHANCED SYNTHETIC DATA GENERATION FOR AUTONOMOUS VEHICLE TRAINING

Project ID – R26-CS-010

Student ID – IT22541284

Name – Yubitha S

Individual Component – Generate realistic, privacy aware Synthetic autonomous vehicle datasets, with security features.

Summary of RP work carried out during the reporting period.

Date	RP Work
16/09/2026	Incorporated supervisor feedback on the research paper draft. Revised the security layer description, strengthened the experimental results section and updated the conclusion.
24/09/2026	Finalised the research paper for submission. Formatted according to target publication guidelines. Confirmed all references are correctly cited and formatted.
27/09/2026	Completed final group thesis submission package. Confirmed all appendices including RBAC policy documentation, hash manifest samples and encryption configuration are included.
02/10/2026	Submitted final group report ahead of the 11 Oct 2026 deadline. Reviewed submission checklist with group members to confirm all components are included and correct.
09/10/2026	Rehearsed viva presentation with group members. Practised responding to technical questions on the cybersecurity layer design, RBAC policy and dataset security outcomes.
14/10/2026	Finalised research paper submission package for the 15 Oct deadline. Completed publication evidence documentation in preparation for the 1 Dec submission milestone.

Date

16th September – 15st October 2026

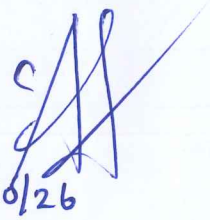
Supervisor's Comments :-

Student progressing well in research project :- Yes / No

If No :-

Project work to be carried out before next evaluation period :-

Code base checked :- Yes / No



Signature :-

Date :- 19/10/26